

Nonlinear inversion

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Nonlinear inversion

Nonlinear inverse problems:

- Seismic FWI: nonlinear in elastic parameters
- EM inversion: nonlinear in electric conductivity (resistivity)
- GravMag inversion: nonlinear in source locations

$$\mathbf{d} = \mathcal{F}(\mathbf{m}) + \mathbf{n} \quad (\mathcal{F}(\mathbf{m}) \neq \mathbf{A}\mathbf{m})$$

Objective function:

$$\begin{aligned} \Phi &= \frac{1}{2}[\mathbf{d} - \mathcal{F}(\mathbf{m})]^T \Sigma_n^{-1} [\mathbf{d} - \mathcal{F}(\mathbf{m})] && \text{Max likelihood} \\ &+ \frac{1}{2}[\mathbf{m} - \mathbf{m}_0]^T \Sigma_m^{-1} [\mathbf{m} - \mathbf{m}_0] && \text{Regularization} \end{aligned}$$

Gradient and Hessian with respect to model parameters

$$\begin{aligned}\frac{\partial \Phi}{\partial \mathbf{m}} &= \mathbf{g}(\mathbf{m}) \\ \frac{\partial^2 \Phi}{\partial \mathbf{m}^2} &= \mathbf{H}(\mathbf{m}) = \frac{\partial \mathbf{g}}{\partial \mathbf{m}}\end{aligned}$$

Find a model update $\Delta \mathbf{m}$ such that

$$\begin{aligned}\mathbf{g}(\mathbf{m} + \Delta \mathbf{m}) &= 0 \\ \mathbf{g}(\mathbf{m} + \Delta \mathbf{m}) &\simeq \mathbf{g}(\mathbf{m}) + \frac{\partial \mathbf{g}}{\partial \mathbf{m}}(\mathbf{m}) \Delta \mathbf{m} = 0\end{aligned}$$

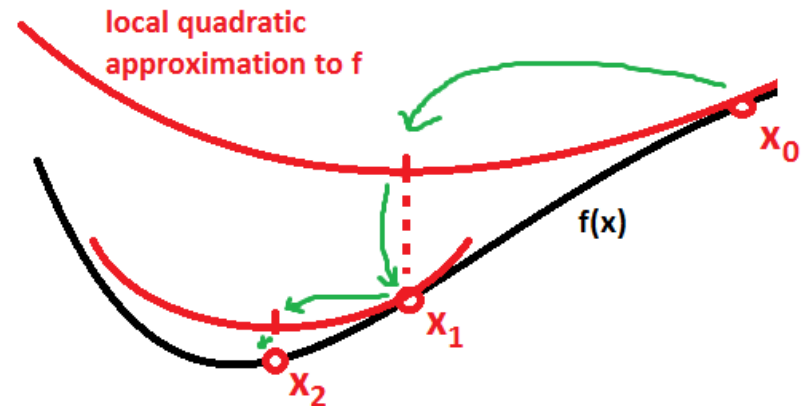
Linear system of equations

$$\mathbf{g} + \mathbf{H} \Delta \mathbf{m} = 0 \quad \Rightarrow \quad \Delta \mathbf{m} = -\mathbf{H}^{-1} \mathbf{g}$$

$$\begin{aligned}\mathbf{g} &= \text{gradient} \\ \mathbf{H} &= \text{Hessian}\end{aligned}$$

Iterative solution:

$$\mathbf{m}^{(n+1)} = \mathbf{m}^{(n)} - \mathbf{H}^{-1} \mathbf{g}^{(n)}$$



Three main challenges:

- Computing the gradient
- Computing the Hessian
- Regularization

Forward modeling (FD/FE) has high compute cost

- Electromagnetics: Solve Maxwell equations
- Seismic FWI: Solve acoustic or elastic wave equation

Gradient and Hessian

Gradient:

$$\begin{aligned}\mathbf{g} &= \frac{\partial \Phi}{\partial \mathbf{m}} = -\frac{\partial \mathcal{F}^T}{\partial \mathbf{m}} \Sigma_n^{-1} \Delta \mathbf{d} + \frac{\partial C}{\partial \mathbf{m}} \\ \Delta \mathbf{d} &= \mathbf{d} - \mathcal{F}(\mathbf{m}) = \text{data residual} \\ C &= \text{regularization}\end{aligned}$$

Hessian:

$$\mathbf{H} = \frac{\partial^2 \Phi}{\partial \mathbf{m}^2} = \frac{\partial \mathcal{F}^T}{\partial \mathbf{m}} \Sigma_n^{-1} \frac{\partial \mathcal{F}}{\partial \mathbf{m}} + \cancel{\frac{\partial^2 \mathcal{F}^T}{\partial \mathbf{m}^2} \Sigma_n^{-1} \Delta \mathbf{d}} + \frac{\partial^2 C}{\partial \mathbf{m}^2}$$

Gauss-Newton: Neglect $\frac{\partial^2 \mathcal{F}}{\partial \mathbf{m}^2}$ term

Jacobian matrix:

$$\mathbf{J} = \frac{\partial \mathcal{F}}{\partial \mathbf{m}}$$

Dimensions of the Jacobi matrix: $N_d \times N_m$

Then

$$\mathbf{g} = \mathbf{J}^H \Sigma_n^{-1} \Delta \mathbf{d} + \frac{\partial C}{\partial \mathbf{m}}$$

Gauss-Newton

$$\mathbf{H} \simeq \mathbf{J}^H \Sigma_n^{-1} \mathbf{J} + \frac{\partial^2 C}{\partial \mathbf{m}^2} \quad (\text{neglecting } \partial^2 \mathcal{F} / \partial \mathbf{m}^2 \text{ terms})$$

Computation of Jacobian matrix requires forward modeling in perturbed models

- 2D: $N_m = N_x N_y \sim 10^4$
- 3D: $N_m = N_x N_y N_z \sim 10^6$
- The naive approach is not feasible with FD and FE forward modeling
- Feasible solution: Use adjoint-state (back-propagation) methods
- For gravity and magnetics it's easier ...

Magnetic inversion – revisited

Magnetics (and gravity):

- The gradient and Hessian can be computed analytically

Magnetic anomaly forward model

$$\mathbf{B}_A(\mathbf{x}) = -\frac{\mu_0}{4\pi} \nabla \nabla \cdot \iiint \frac{\mathbf{M}(\mathbf{x}')}{|\mathbf{x}' - \mathbf{x}|} dV'$$

$$\mathbf{M} = \mathbf{M}_R + \mathbf{M}_I = \mathbf{M}_R + \chi \mathbf{H}$$

Approximation: Assume $\mathbf{M}$ is independent of z or average over z

$$\mathbf{M}(x, y) = \frac{1}{z_2 - z_1} \int_{z_1}^{z_2} \mathbf{M}(x, y, z) dz$$

Then we can perform the integration over z in the equation for $\mathbf{B}_A$

$$\int_{z_1}^{z_2} \frac{1}{r} dz' = \ln[(z' - z) + r] \Big|_{z_1}^{z_2} = F(z_2) - F(z_1)$$

Magnetic inversion – revisited

Forward model

$$B_A(\mathbf{x}) \simeq \mathbf{t} \cdot \mathbf{B}_A(\mathbf{x}) = \frac{\mu_0}{4\pi} \sum_{i=1}^3 \sum_{j=1}^3 \iint t_i A_{ij}(\mathbf{x}, \mathbf{x}') M_j(x', y') dx' dy'$$

$$A_{ij} = \frac{\partial^2 F(z_2)}{\partial x_i \partial x_j} - \frac{\partial^2 F(z_1)}{\partial x_i \partial x_j}$$

Discretize and prepare for inversion

$$B_A(\mathbf{x}_i) = \sum_j Q_M(\mathbf{x}_i, \mathbf{x}'_j) M(\mathbf{x}'_j)$$

$$Q_M(\mathbf{x}_i, \mathbf{x}'_j) = \frac{\mu_0}{4\pi} \Delta S'_j \sum_{k=1}^3 \sum_{l=1}^3 t_k A_{kl}(\mathbf{x}_i, \mathbf{x}'_j) e_l,$$

$$\mathbf{M} = M \mathbf{e}$$

$$M_j = M e_j$$

$$A_{ij} = \frac{\partial^2 F(z_2)}{\partial x_i \partial x_j} - \frac{\partial^2 F(z_1)}{\partial x_i \partial x_j}$$

$$\begin{aligned} \frac{\partial^2 F(z')}{\partial x_i \partial x_j} = & -\frac{1}{[(z' - z) + r]^2} \left[\delta_{i3} + \frac{(x'_i - x_i)}{r} \right] \left[\delta_{j3} + \frac{(x'_j - x_j)}{r} \right] \\ & + \frac{1}{[(z' - z) + r]} \left[\frac{\delta_{ij}}{r} - \frac{(x'_i - x_i)(x'_j - x_j)}{r^3} \right] \end{aligned}$$

Linear magnetic inversion for M

Linear inversion; standard form

$$\mathbf{d} = \mathbf{Lm} + \mathbf{n}$$

$$\Phi = \frac{1}{2} [\mathbf{d} - \mathbf{Lm}]^T [\mathbf{d} - \mathbf{Lm}] + \lambda \frac{1}{2} \mathbf{m}^T \mathbf{D}^T \mathbf{D} \mathbf{m}$$

$$d_i = B_A(\mathbf{x}_i)$$

$$m_j = M(\mathbf{x}'_j)$$

$$L_{ij} = Q(\mathbf{x}_i, \mathbf{x}'_j)$$

Marquardt-Levenberg solution

$$\mathbf{m} = [\mathbf{L}^T \mathbf{L} + \lambda \mathbf{D}^T \mathbf{D}]^{-1} \mathbf{L}^T \mathbf{d}$$

$$\mathbf{D}^T \mathbf{D} = \text{diag}(\mathbf{L}^T \mathbf{L})$$

Non-linear magnetic inversion for $\mathbf{M}$ and $\mathbf{z}_2$

Non-linear inversion; standard form

$$\mathbf{d} = \mathbf{f}(\mathbf{m}) + \mathbf{n}$$

$$\Phi = \frac{1}{2} [\mathbf{d} - \mathbf{f}(\mathbf{m})]^T [\mathbf{d} - \mathbf{f}(\mathbf{m})] + \lambda \frac{1}{2} \mathbf{m}^T \mathbf{D}^T \mathbf{D} \mathbf{m}$$

Model parameters

$$\mathbf{m} = [\mathbf{M}, \mathbf{z}_2]^T$$

Gradient and Hessian

$$\mathbf{g}(\mathbf{m}) = -\mathbf{J}^T [\mathbf{d} - \mathbf{f}(\mathbf{m})] + \lambda \mathbf{D}^T \mathbf{D} \mathbf{m}$$

$$\mathbf{H} \simeq \mathbf{J}^T \mathbf{J} + \lambda \mathbf{D}^T \mathbf{D}$$

The physical forward model is the same as before, but $\mathbf{m}$ is different

$\mathbf{M}$ = magnetization

$\mathbf{z}_2$ = base of source layer

Non-linear magnetic inversion Jacobi matrix

Jacobi matrix

$$\mathbf{J} = \frac{\partial \mathbf{f}}{\partial \mathbf{m}} = [\mathbf{L}, \mathbf{K}]$$

$$L_{ij} = \frac{\partial B_A(\mathbf{x}_i)}{\partial M(\mathbf{x}'_j)} = Q_M(\mathbf{x}_i, \mathbf{x}'_j) \quad (\text{as before})$$

$$K_{ij} = \frac{\partial B_A(\mathbf{x}_i)}{\partial z_2(\mathbf{x}'_j)} = \frac{\partial Q_M(\mathbf{x}_i, \mathbf{x}'_j)}{\partial z_2(\mathbf{x}'_j)} M(\mathbf{x}'_j) \quad (\text{the non-linear part})$$

Reminding ourselves of the forward model

$$\mathbf{f}(\mathbf{m}) = B_A(\mathbf{x}_i) = \sum_j Q_M(\mathbf{x}_i, \mathbf{x}'_j) M(\mathbf{x}'_j)$$
$$Q_M(\mathbf{x}_i, \mathbf{x}'_j) = \frac{\mu_0}{4\pi} \Delta S'_j \sum_{k=1}^3 \sum_{l=1}^3 t_k A_{kl}(\mathbf{x}_i, \mathbf{x}'_j) e_l,$$

Working out the details

$$\frac{\partial Q_M}{\partial z_2} = \frac{\mu_0}{4\pi} \Delta S'_j \sum_{k=1}^3 \sum_{l=1}^3 t_k \left[\frac{\partial}{\partial z_2} \frac{\partial^2 F(z_2)}{\partial x_k \partial x_l} \right] e_l$$

$$A_{ij} = \frac{\partial^2 F(z_2)}{\partial x_i \partial x_j} - \frac{\partial^2 F(z_1)}{\partial x_i \partial x_j}$$

$$F(z_2) - F(z_1) = \ln[(z' - z) + r] \Big|_{z_1}^{z_2} = \int_{z_1}^{z_2} \frac{1}{r} dz'$$

This brings us back to the well-known magnetostatic equation

$$\frac{\partial}{\partial z_2} \frac{\partial^2 F(z_2)}{\partial x_i \partial x_j} = -\frac{\delta_{ij}}{r_{ij}^3} + 3 \frac{(x'_i - x_i)(x'_j - x_j)}{r_{ij}^5}$$

Iterativ solution – model update

Gradient and Hessian (for current model $\mathbf{m}_k$)

$$\mathbf{g}(\mathbf{m}_k) = -\mathbf{J}^T[\mathbf{d} - \mathbf{f}(\mathbf{m}_k)] + \lambda \mathbf{D}^T \mathbf{D} \mathbf{m}_k$$

$$\mathbf{H} \simeq \mathbf{J}^T \mathbf{J} + \lambda \mathbf{D}^T \mathbf{D}$$

Marquardt-Levenberg

$$\mathbf{D}^T \mathbf{D} = \text{diag}(\mathbf{J}^T \mathbf{J})$$

Model update

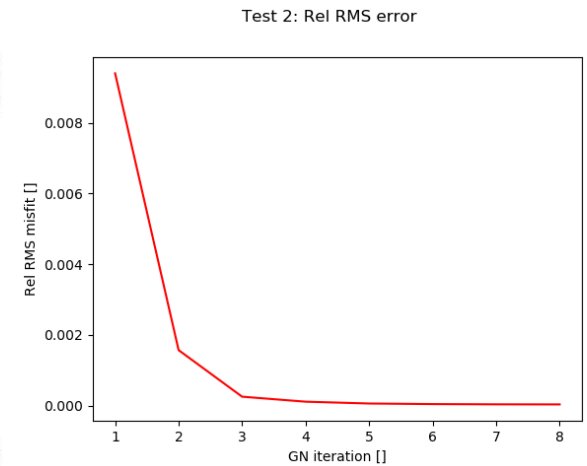
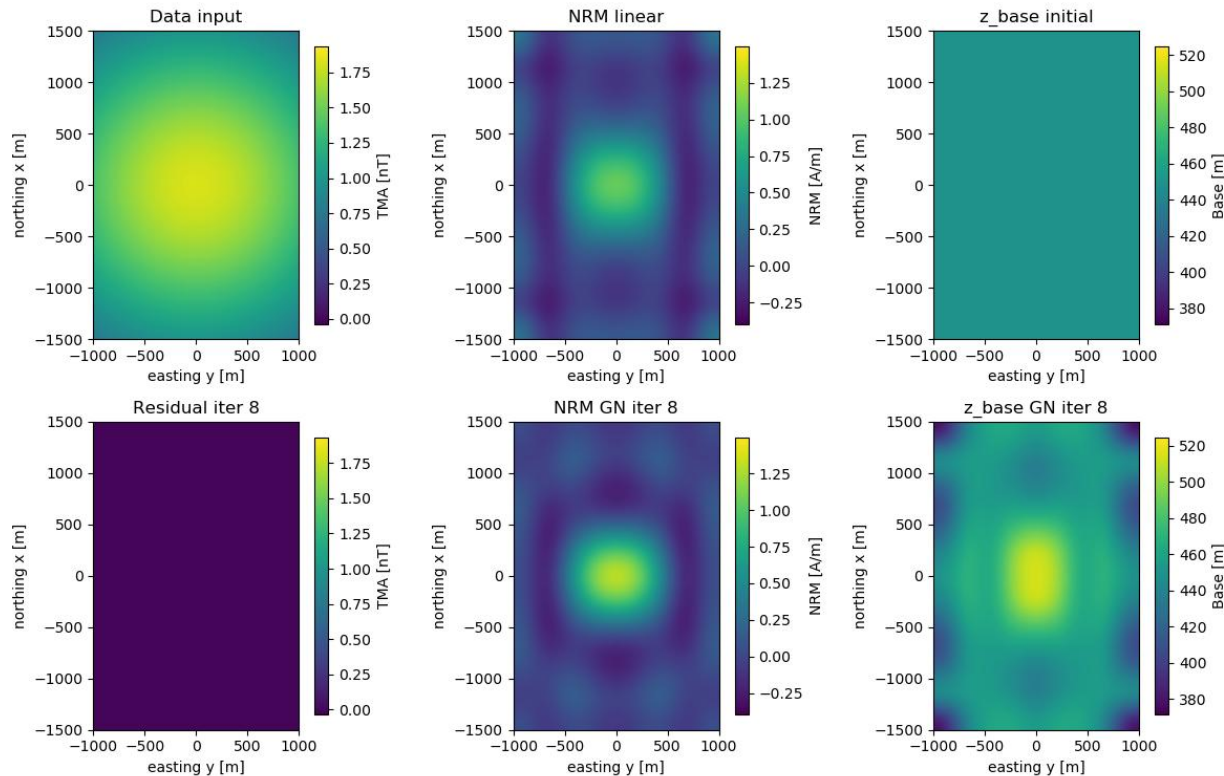
$$\Delta \mathbf{m}_k = [\mathbf{J}^T \mathbf{J} + \lambda \mathbf{D}^T \mathbf{D}]^{-1} \mathbf{J}^T [\mathbf{d} - \mathbf{f}(\mathbf{m}_k)]$$

$$\mathbf{m}_{k+1} = \mathbf{m}_k + \Delta \mathbf{m}_k$$

Non-linear inversion for NRM and base magnetic source layer

Measured 3000m above top magnetic source

Test 2: Inversion: $dx=dy=250.0$



Gradient by back-propagation (adjoint state)

Toy-theory for EM inversion

Consider EM, isotropic case.

Objective function in frequency domain:

$$\Phi = \frac{1}{4\pi} \int d\omega \int_R d^2 R |\Delta E(\mathbf{x}_r, \omega)|^2 + \text{regularization}$$

$$\Delta E = E_x(\mathbf{x}_r, \omega) - \mathcal{F}_x(\mathbf{x}_r, \omega, \sigma(\mathbf{x})) = \text{data misfit}$$

$$\mathcal{F}_x = \text{forward modeling (solving Maxwell equations)}$$

$$\mathbf{x}_r = \text{receiver positions}$$

$$R = \text{recording surface}$$

$$\sigma = 1/\rho = \text{conductivity}$$

Maxwell equations - isotropic

Maxwell:

$$\nabla \times \mathbf{E} = -\mu_0 \partial_t \mathbf{H}$$

$$\nabla \times \mathbf{H} = \mathbf{j}^E$$

Ohm's law:

$$\mathbf{j}^E = \sigma \mathbf{E}$$

Vector manipulation:

$$\nabla \times (\nabla \times \mathbf{E}) = -\mu_0 \sigma \partial_t \mathbf{E}$$

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}$$

$$\nabla \cdot \mathbf{E} = 0$$

$$(\text{Anisotropic: } \nabla \cdot \mathbf{j} = \partial_i(\sigma_{ij} E_j) = 0)$$

Diffusive "wave equation" (isotropic):

$$\partial_j \partial_j E_i - \mu_0 \sigma \partial_t E_i = 0$$

$$\partial_j \partial_j E_i + i\omega \mu_0 \sigma E_i = 0$$

Gradient wrt $\sigma(\mathbf{x})$

$$\begin{aligned} g(\mathbf{x}) &= \frac{\partial \Phi}{\partial \sigma(\mathbf{x})} \\ &= \frac{-1}{2\pi} \int d\omega \int d^2 R \left[\frac{\partial \mathcal{F}^*(\mathbf{x}_r, \omega, \sigma(\mathbf{x}))}{\partial \sigma(\mathbf{x})} \Delta E(\mathbf{x}_r, \omega) \right] \\ &= \frac{-1}{2\pi} \int d\omega \int d^2 R [K^*(\mathbf{x}_r, \omega, \sigma(\mathbf{x})) \Delta E(\mathbf{x}_r, \omega)] \\ K &= \frac{\partial \mathcal{F}}{\partial \sigma} = \text{Frechet kernel} \end{aligned}$$

Perturb σ to compute K

$$\begin{aligned} \sigma(\mathbf{x}) &= \sigma_0(\mathbf{x}) + \delta\sigma(\mathbf{x}) \\ \sigma_0 &= \text{background conductivity} \\ \delta\sigma &= \text{perturbation} \end{aligned}$$

Helmholtz equation for Green's functions

$$(\nabla^2 + i\omega\mu_0\sigma_0)G_0(\mathbf{x}, \mathbf{x}_r) = \delta(\mathbf{x} - \mathbf{x}_r) \quad (1)$$

$$(\nabla^2 + i\omega\mu_0\sigma)G(\mathbf{x}, \mathbf{x}_s) = \delta(\mathbf{x} - \mathbf{x}_s) \quad (2)$$

Multiply (1) by G , (2) by G_0 , subtract and integrate

$$\begin{aligned} & \int d^3x [G(\mathbf{x}, \mathbf{x}_s)\delta(\mathbf{x} - \mathbf{x}_r) - G_0(\mathbf{x}, \mathbf{x}_r)\delta(\mathbf{x} - \mathbf{x}_s)] \\ &= \int d^3x [G(\mathbf{x}, \mathbf{x}_s)\nabla^2 G_0(\mathbf{x}, \mathbf{x}_r) - G_0(\mathbf{x}, \mathbf{x}_r)\nabla^2 G(\mathbf{x}, \mathbf{x}_s)] \\ & \quad + i\omega\mu_0 \int d^3x [\sigma(\mathbf{x}) - \sigma_0(\mathbf{x})]G(\mathbf{x}, \mathbf{x}_s)G_0(\mathbf{x}, \mathbf{x}_r) \end{aligned}$$

Re-write ∇^2 -terms using Green's theorem

$$G(\mathbf{x}_r, \mathbf{x}_s) - G_0(\mathbf{x}_s, \mathbf{x}_r) = i\omega\mu_0 \int d^3x \delta\sigma(\mathbf{x}) G(\mathbf{x}, \mathbf{x}_s) G_0(\mathbf{x}, \mathbf{x}_r) \\ + \int dS \mathbf{n} \cdot [G(\mathbf{x}, \mathbf{x}_s) \nabla G_0(\mathbf{x}, \mathbf{x}_r) - G_0(\mathbf{x}, \mathbf{x}_r) \nabla G(\mathbf{x}, \mathbf{x}_s)]$$

Get rid of the surface integral by Sommerfeldt radiation condition

Lippmann-Schwinger equation

$$G(\mathbf{x}_r, \mathbf{x}_s) = G_0(\mathbf{x}_r, \mathbf{x}_s) + i\omega\mu_0 \int d^3x G(\mathbf{x}, \mathbf{x}_s) \delta\sigma(\mathbf{x}) G_0(\mathbf{x}, \mathbf{x}_r)$$

- Lippmann-Schwinger is an integral equation for G

Born approximation:

$$G(\mathbf{x}_r, \mathbf{x}_s) = G_0(\mathbf{x}_s, \mathbf{x}_r) + i\omega\mu_0 \int d^3x G_0(\mathbf{x}, \mathbf{x}_s) \delta\sigma(\mathbf{x}) G_0(\mathbf{x}, \mathbf{x}_r)$$

- Single-scattering approximation

Frechet derivative

$$\delta G(\mathbf{x}_r, \mathbf{x}_s) = i\omega\mu_0 \int d^3x G(\mathbf{x}, \mathbf{x}_s) \delta\sigma(\mathbf{x}) G_0(\mathbf{x}, \mathbf{x}_r)$$

$$\frac{\delta G}{\delta\sigma} = G_0(\mathbf{x}, \mathbf{x}_s) G_0(\mathbf{x}, \mathbf{x}_r)$$

Frechet derivative (functional derivative) defined

$$S(f) = \int L(x, f(x), f'(x)) dx$$

$$\frac{\delta S}{\delta f} = \frac{\partial L}{\partial f} - \frac{d}{dx} \frac{\partial L}{\partial f'}$$

Account for the dipole source current

$$\begin{aligned} K(\mathbf{x}_r, \mathbf{x}) &= i\omega\mu_0 G_0(\mathbf{x}_r, \mathbf{x}) \int d^3x_s G_0(\mathbf{x}, \mathbf{x}_s) J(\mathbf{x}_s) \\ &= i\omega\mu_0 G_0(\mathbf{x}_r, \mathbf{x}) E_0(\mathbf{x}) \end{aligned}$$

J = source current density

E_0 = incident field from the source

Substitute for K^* in equation for g

$$g(\mathbf{x}) = \frac{1}{2\pi} \int d\omega i\omega\mu_0 E_0^*(\mathbf{x}) \underbrace{\int d^2R G_0^*(\mathbf{x}, \mathbf{x}_r) \Delta E(\mathbf{x}_r)}_{\text{Back-propagation}}$$

Just to remind ourselves what we're working on:

Gradient wrt $\sigma(\mathbf{x})$

$$\begin{aligned} g(\mathbf{x}) &= \frac{\partial \Phi}{\partial \sigma(\mathbf{x})} \\ &= \frac{-1}{2\pi} \int d\omega \int d^2 R \left[\frac{\partial \mathcal{F}^*(\mathbf{x}_r, \omega, \sigma(\mathbf{x}))}{\partial \sigma(\mathbf{x})} \Delta E(\mathbf{x}_r, \omega) \right] \\ &= \frac{-1}{2\pi} \int d\omega \int d^2 R [K^*(\mathbf{x}_r, \omega, \sigma(\mathbf{x})) \Delta E(\mathbf{x}_r, \omega)] \\ K &= \frac{\partial \mathcal{F}}{\partial \sigma} = \text{Frechet kernel} \end{aligned}$$

Relation to migration

$$D(\mathbf{x}) = E_0(\mathbf{x})$$

Forward modeling

$$U(\mathbf{x}) = \int d^2 R G_0^*(\mathbf{x}, \mathbf{x}_r) \Delta E(\mathbf{x}_r)$$

Back-propagation

$$g(\mathbf{x}) = \frac{\mu_0}{2\pi} \int d\omega i\omega U(\mathbf{x}) D^*(\mathbf{x})$$

Gradient

$$\hat{g}(\mathbf{x}) = \frac{\mu_0}{2\pi} \frac{\int d\omega i\omega U(\mathbf{x}) D^*(\mathbf{x})}{\int d\omega D(\mathbf{x}) D^*(\mathbf{x})}$$

Preconditioned gradient

- Claerbout's U/D principle
- Gradient computation is migration

Steepest decent, line search

Preconditioned gradient

$$\begin{aligned}\hat{g}(\mathbf{x}) &= B(\mathbf{x})g(\mathbf{x}) \\ 1/B(\mathbf{x}) &= \int d\omega E_0(\mathbf{x})E_0^*(\mathbf{x})\end{aligned}$$

Steepest-decent model update

$$\begin{aligned}\sigma^{n+1}(\mathbf{x}) &= \sigma^n(\mathbf{x}) - \alpha\hat{g}(\mathbf{x}) \\ \alpha &= \text{steplength}\end{aligned}$$

- $\alpha B(\mathbf{x})$ is the simplest approximation to the Hessian

~~Our next task is to compute the steplength!~~

Back tracking:

If

$$\Phi(\mathbf{m}^{(n)} - \alpha \hat{g}^{(n)}) > \Phi(\mathbf{m}^{(n)})$$

try

$$\mathbf{m}^{(n+1)} = \mathbf{m}^{(n)} - \frac{\alpha}{2} \hat{g}^{(n)}$$

and so on ...

Steepest decent - summary

$$\sigma^{(n+1)}(\mathbf{x}) = \sigma^{(n)}(\mathbf{x}) - \alpha \hat{g}^{(n)}(\mathbf{x})$$

$$\hat{g}^{(n)}(\mathbf{x}) = B(\mathbf{x})g^{(n)}(\mathbf{x})$$

$$\alpha = \frac{2\pi \int d^3x g(\mathbf{x})\hat{g}(\mathbf{x})}{\int d\omega \int dR \left| \frac{\Delta \tilde{E}(\mathbf{x}_r)}{\beta} \right|^2}$$

The model update depends only on g at iteration n

Why our toy theory doesn't work in practise

- Too simple forward model (isotropic)
- No data weights
- No regularization
- Steepest decent goes straight to the first local minumum

Conjugate gradient

$$\sigma^{(n+1)}(\mathbf{x}) = \sigma^{(n)}(\mathbf{x}) - \alpha \hat{g}_c^{(n)}(\mathbf{x})$$

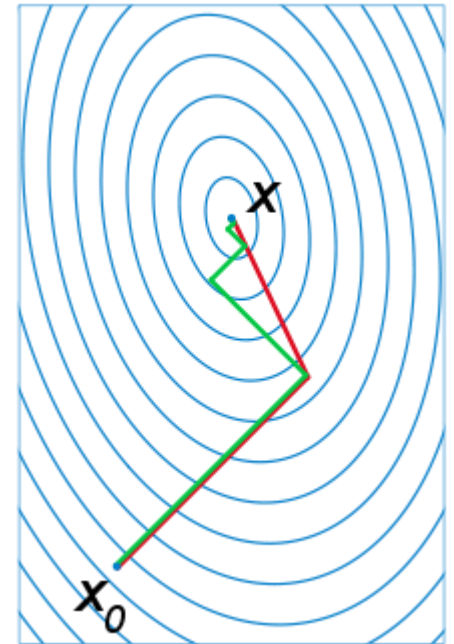
$$\hat{g}_c^{(n)}(\mathbf{x}) = \hat{g}^{(n)}(\mathbf{x}) + \gamma \hat{g}_c^{(n-1)}(\mathbf{x})$$

$$\gamma = \frac{\int d^3x (g^{(n)}(\mathbf{x}) - g^{(n-1)}(\mathbf{x})) \hat{g}^{(n)}(\mathbf{x})}{\int d^3x \hat{g}^{(n)}(\mathbf{x}) g^{(n)}(\mathbf{x})}$$

Keeps memory of previous gradient directions.

First iteration equals steepest decent:

$$\hat{g}_c^{(1)}(\mathbf{x}) = \hat{g}^{(1)}(\mathbf{x})$$



Steepest decent vs
conjugate gradient
(from Wikipedia)

BFGS approximation

BFGS approximation

- Update (inverse) Hessian based on (inverse) Hessians from L previous iterations
- Needs to store $n \times n$ matrices

$$\mathbf{H}^{(k+1)} = \mathbf{H}^{(k)} + \alpha \mathbf{u} \mathbf{u}^T + \beta \mathbf{v} \mathbf{v}^T$$

$$\mathbf{u} = \Delta \mathbf{g}^{(k)}$$

$$\mathbf{v} = \mathbf{H}^{(k)} \Delta \mathbf{m}^{(k)}$$

L-BFGS approximation

- Avoid creating and storing the (inverse) Hessian matrix
- Depends only on the gradients and model updates from last L iterations (typically $L \sim 10$)
- Computes the update direction $-\mathbf{H}^{-1} \mathbf{g}$ directly
- Start from a simple (diagonal) $\mathbf{H}_0$
- Matrix free
- Popular in non-linear inversion (FWI and CSEM)

Time domain – Reverse time migration

$$g(\mathbf{x}, \mathbf{x}_s) = -\mu_0 \int_{t_1}^{t_2} dt \partial_t E^{FW}(\mathbf{x}, \mathbf{x}_s, t) E^{BW}(\mathbf{x}, \mathbf{x}_s, t)$$

$$B^{-1}(\mathbf{x}, \mathbf{x}_s) = \int_{t_1}^{t_2} dt E^{FW}(\mathbf{x}, \mathbf{x}_s, t) E^{FW}(\mathbf{x}, \mathbf{x}_s, t)$$

Seismic:

$$g(\mathbf{x}, \mathbf{x}_s) = \frac{1}{v^2} \int_{t_1}^{t_2} dt \partial_t P^{FW}(\mathbf{x}, \mathbf{x}_s, t) \partial_t P^{BW}(\mathbf{x}, \mathbf{x}_s, t)$$

Parseval's theorem

$$\int x(t) y^*(t) dt = \int X(\omega) Y^*(\omega) d\omega$$

Correlation can be done in time or frequency

Vintage 2D prestack reverse-time migration

